# Till Freihaut

PhD Student · Computer Science

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## Education

**University of Zurich & ETH AI Center** — *Zurich*

PhD Computer Science — *05/2024 - present*

- Advisor: Prof. Dr. Giorgia Ramponi
- Research Group: Autonomous Learning and Predictive Intelligence Lab
- Research Topics: Multi-agent Reinforcement Learning, Imitation Learning
- ETH AI Center: Associate PhD

**University of Mannheim** — *Mannheim*

MS Mathematics, 1.0 with Honors — *09/21 - 03/24*

- Thesis: Importance Sampling in Offline Contextual Bandits
- Advisors: Claire Vernade and Leif Döring
- Focus: Mathematics in Artificial Intelligence, in particular Reinforcement Learning
- Honors: Finished with best possible grade

**University of Mannheim** — *Mannheim*

BS Mathematics with a Minor in Business, 1.8 — *08/18 - 08/21*

- Focus: Probability Theory and Statistics
- Thesis: Stochastic Approximation in Reinforcement Learning
- Supervisor: Prof. Dr. Leif Döring

## Professional Experience

**Point 72** — *New York*

Machine Learning Researcher — *05/2026 - 08/2026*

- Internal Alpha Catching. Applied Machine Learning project for financial data.

**ETH AI Center** — *Zurich*

Associate PhD student — *05/2024 - present*

- Collaborating with fellow PhD students and participating in industry networking events.

**adago (formerly adacon)** — *Mannheim*

Data Scientist — *01/2022 - 05/2024*

- Collaborated with research scientists to develop Reinforcement Learning algorithms for supply-chain systems using Python and stable-baselines3.

**University of Mannheim** — *Mannheim*

Graduate Reinforcement Learning Research Assist — *10/2021 - 09/2022*

- Supported reinforcement learning research, instructed a Python course for doctoral students, and developed programming exercises for a Reinforcement Learning course.

**Deutsche Bank** — *Berlin*

Quantitative Risk Management Intern — *08/2021 - 12/2021*

- Automated risk analysis for trading products using Python, optimized code performance, and assisted in deploying a data analysis tool.

**University of Mannheim** — *Mannheim*

Undergraduate Teaching Assistant — *08/2020 - 02/2021*

- Led tutorials for over 20 students in analysis, supported their understanding of course concepts, and graded assignments and exams.

## Publications

### Conferences

Luca Viano*, **Till Freihaut***, Emanuele Nevali, Volkan Cevher, Matthieu Geist, Giorgia Ramponi. Multi-agent imitation learning with function approximation: Linear Markov games and beyond, ICML2026,Paper Link

**Till Freihaut**, Giorgia Ramponi. On Feasible Rewards in Multi-Agent Inverse Reinforcement Learning, **Spotlight** at NeurIPS 2025 (Top 3% of papers), Paper Link

**Till Freihaut***, Luca Viano*, Volkan Cevher, Matthieu Geist, Giorgia Ramponi. Learning equilibria from Data: Provably-efficient Multi-Agent Imitation Learning, NeurIPS 2025 & **Spotlight** at RLBrew Workshop at RLC, Paper Link

**Till Freihaut***, Luca Viano*, Emanuele Nevali,Volkan Cevher, Matthieu Geist, Giorgia Ramponi. Rate optimal learning equilibria from data, AISTATS2026,Paper Link

### Workshops

Simon Weissmann*, **Till Freihaut***, Claire Vernade, Giorgia Ramponi, Claire Vernade, Leif Döring. Barycenter Policy Design for Multiple Policy Evaluation, EWRL2025, Paper Link Victor Villin, **Till Freihaut***, Andreas Schlaginhaufen*, Christos Dimitrakakis, Maryam Kamgarpour, Giorgia Ramponi. Inferring Transferable Rewards via Active Inverse Reinforcement Learning, Oral at EWRL2026, Paper Link

### Preprint

Victor Villin, **Till Freihaut***, Andreas Schlaginhaufen*, Christos Dimitrakakis, Maryam Kamgarpour, Giorgia Ramponi. Inferring Transferable Rewards via Active Inverse Reinforcement Learning, Paper Link

### Packages

Benedikt Geier, Borui Niklas Zhu, Moritz Kern, Manuel J. Hentschel, Kiril Dik, Moritz Lauff, Tim Glockner, Adrian Heppeler, Nadine Tampe, Leonardo Vela, **Till Freihaut**, Niclas Lietzow, Helene Peter, Martin Schlather, Yiqi Li. Package. 2020. DistributionFitR, CRAN

## Awards & Honors

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| 2026 | **UZH PhD Candoc Grant**, University of Zurich |
| 2024 | **Best Student Recognition**, Master Mathematics, University of Mannheim |
| 2024 | **Werner Oettli Prize (Best Master Thesis)**, Master Mathematics, University of Mannheim |
| 2018 | **Best High school Graduate in Mathematics** , Deutsche Mathematiker Vereinigung (German Mathematics Association) |

## Teaching Experience

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| Fall 2026 | **Reinforcement Learning**, Teaching Assistant | *Zurich* |
| Fall 2025 | **Reinforcement Learning**, Head Teaching Assistant | *Zurich* |
| Fall 2024 | **Reinforcement Learning**, Head Teaching Assistant | *Zurich* |
| Fall 2020 | **Analysis I**, Teaching Assistant | *Mannheim* |

## Student Supervision

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| 2024-now | **Master students at ETH, University of Zurich and EPFL**, Various topics in Reinforcement Learning | *Zurich* |

## Scientific Activities

### Organizing

**NeurIPS Workshop, 2025,** Co-organizing the Second Edition of Aligning Reinforcement Learning Experimentalists and Theorists (ARLET); Organizing Committee: Marco Mussi, **Till Freihaut**, Antoine Moulin, Giorgia Ramponi, Dirk van der Hoeven, Alberto Maria Metelli, Felix Berkenkamp, Francesco Trovò , Csaba Szepesvári.

## Reviewer

ICML (2026), AISTATS (2026, 2027), EWRL (2025, 2026), FLLMPT and EXAIT Workshop ICML (2025, 2026), ARLET Workshop ICML(2024)